COMP721 SEMESTER PROJECT

SUBMISSION GUIDE AND ASSESSMENT MILESTONES

- The semester project will be completed **in groups of four (4) students**. The project has been divided into **three submission phases** to ensure continuous progress throughout the semester.
- Each submission is compulsory. Failure to submit any phase may result in a mark of zero for that phase.
- All submissions must be made via **Moodle** before the specified deadline.

Project Timeline:

Phase	Submission	Deadline	Marks
Phase 1	Project Proposal	**14 August 2026 (23:59)**	10
Phase 2	Progress Report + Partial Code	**11 September 2026 (23:59)**	30
Phase 3	Final Submission	**17 October 2026 (23:59)**	60
Presentation	Oral Presentation	**26 October 2026 (09:00–12:00)**	50

Total Project Marks: 150

PHASE 1: PROJECT PROPOSAL

Submission Deadline

Date: 14 August 2026

Time: 23:59

Marks: 10 Marks

Objective: The objective of this phase is to ensure that each group has a clear understanding of the project requirements and has developed an implementation plan before beginning the technical work.

Documents to Submit

1. Project Proposal (MS Word and PDF)

Maximum length:

3–5 pages

Proposal Structure

Cover Page

- Module Code
- Module Name
- Project Title
- Group Number
- Student Names
- Student Numbers

1. Introduction

Briefly introduce cardiovascular disease prediction and explain why it is an important machine learning application.

2. Problem Statement

Clearly describe the problem that the project aims to solve.

3. Objectives

State the overall objective and specific objectives of the project.

4. Dataset Description

Provide:

- Dataset source
- Number of records
- Number of features
- Target variable
- Brief description of each feature

5. Proposed Methodology

Briefly describe:

- Data preprocessing
- Feature engineering
- Feature selection
- Machine learning models
- Ensemble method
- Explainability
- Statistical analysis

6. Work Plan

Provide a timeline showing how the group plans to complete the project.

7. References

Use IEEE or APA referencing style.

PHASE 2: PROGRESS REPORT (APPROXIMATELY 50% COMPLETION)

Submission Deadline

Date: 11 September 2026

Time: 23:59

Marks: 30 Marks

Objective: At this stage, groups are expected to have completed approximately **50% of the project**. The aim is to assess progress, identify potential problems early, and provide constructive feedback before the final submission.

Documents to Submit

1. Progress Report (MS Word and PDF)

Maximum length: Approximately **10–15 pages**

Progress Report Structure

Cover Page

Include

- Group Number
- Student Names
- Student Numbers

1. Introduction

Updated introduction.

2. Literature Review

A brief review of related work.

Expected: Approximately **5–10 recent journal/conference papers**.

3. Dataset Description

Include

- Dataset summary
- Feature descriptions
- Target variable

4. Data Preprocessing

Discuss”

- Missing values
- Duplicate removal
- Outlier detection
- Data cleaning
- Feature scaling
- Encoding

Include supporting figures where appropriate.

5. Exploratory Data Analysis

Include

- Descriptive statistics
- Correlation analysis
- Histograms
- Boxplots
- Scatter plots
- Correlation heatmap

Provide discussion of observations.

6. Feature Engineering

Present all engineered features.

Explain

- why they were created,
- how they were created,
- expected benefits.

7. Feature Selection

Present

- feature selection method,
- selected features,
- justification.

8. Preliminary Results

Present any initial experimental results obtained so far.

9. Remaining Work

Describe what remains to be completed before the final submission.

10. Individual Contributions

Each group must include a contribution table.

Example:

Student	Contribution (%)	Responsibilities
Student A	30	Data preprocessing, Feature Engineering
Student B	25	Model Development
Student C	25	SHAP Analysis
Student D	20	Report Writing

The percentages **must total 100%**.

Additional Submission:

Partial Python Code: Submit the current implementation.

The submitted code should include, where applicable:

- Data loading
- Data preprocessing
- EDA
- Feature engineering
- Feature selection
- Any completed model implementation

PHASE 3: FINAL SUBMISSION

Submission Deadline

Date: 17 October 2026

Time: 23:59

Marks: 60 Marks

Objective: Groups are expected to submit the completed project.

Documents to Submit

1. Final Technical Report (MS Word and PDF)

Approximately **20–30 pages,** excluding appendices.

Report Structure

1. Title Page
2. Abstract
3. Introduction
4. Literature Review
5. Methodology

    - Dataset
    - Preprocessing
    - Feature Engineering
    - Feature Selection
    - Model Development
    - Hyperparameter Optimization
    - Cross-validation Strategy
    - Ensemble Learning
    - Explainable AI
    - Statistical Analysis

6. Experimental Setup
7. Results & Discussion
8. Conclusion
9. References
10. Appendices (if applicable)

2. Complete Python Source Code

The submitted code should include:

- Data loading
- Preprocessing
- Exploratory Data Analysis
- Feature Engineering
- Feature Selection
- Hyperparameter Optimization
- Decision Tree
- Support Vector Machine
- Ensemble Learning
- Cross-validation
- Final Testing
- SHAP Analysis
- Statistical Analysis

The code should be well organized, fully commented, and executable.

3. Google Colab Notebook

Submit the complete notebook.

4. Presentation Slides: PowerPoint or PDF.

5. Individual Contribution Statement

Include

- percentage contribution,
- responsibilities,
- tasks completed.

The percentages **must total 100%**.

Final Presentation

Date: 19 October 2026

Time: 09:00–12:00

Venue: Microsoft Teams

Marks: 50 Marks

Presentation Guidelines

- The presentation will take place via **Microsoft Teams** using the meeting link provided on Moodle/Blackboard.
- All groups must be connected by **09:00 AM**, regardless of their allocated presentation slot.
- Each group will be allocated **12 minutes**, consisting of:
    - **10 minutes** for the presentation
    - **2 minutes** for questions and answers
- **All four group members are expected to participate in the presentation**, and each member should contribute meaningfully.
- All presenters must keep their **cameras switched on** throughout their presentation.

Question-and-Answer Protocol

- Each presenting group will be asked **two to three questions**.
- Questions may come from:
    - Other student groups, and/or
    - The lecturer.
- The groups responsible for asking questions will be announced after each presentation.
- Every group should prepare **at least two to three thoughtful questions**, as each group will be required to ask questions during multiple presentations.
- Responses should be provided by members of the presenting group, demonstrating collective understanding of the project.

Presentation Logistics

- If a group finishes early, the next scheduled group should be prepared to begin immediately.
- Groups that are not ready when called may forfeit their presentation opportunity and incur mark penalties.
- Before the presentation day, groups should ensure that:
    - their Microsoft Teams screen-sharing permissions are functioning correctly,
    - presentation slides open properly, and
    - all presenters have tested their microphones, cameras, and internet connectivity.

Presentation Assessment Criteria (50 Marks)

Criterion	Marks
Background, problem statement, and objectives	5
Methodology and technical implementation	5
Data preprocessing, feature engineering, and feature selection	5
Machine learning models, ensemble, and experimental design	5
Results, SHAP analysis, and statistical evaluation	6
Quality of presentation, organisation, and visual aids	4
Responses to questions	20

Total Project Assessment

Component	Marks
Phase 1 – Project Proposal	10
Phase 2 – Progress Report & Partial Code	30
Phase 3 – Final Technical Report & Complete Submission	60
Final Presentation	50
Total	**150**